

9/9/2026

The law of quadratic reciprocity

Thm (Quadratic reciprocity) Let p, q be distinct primes. Then

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}.$$

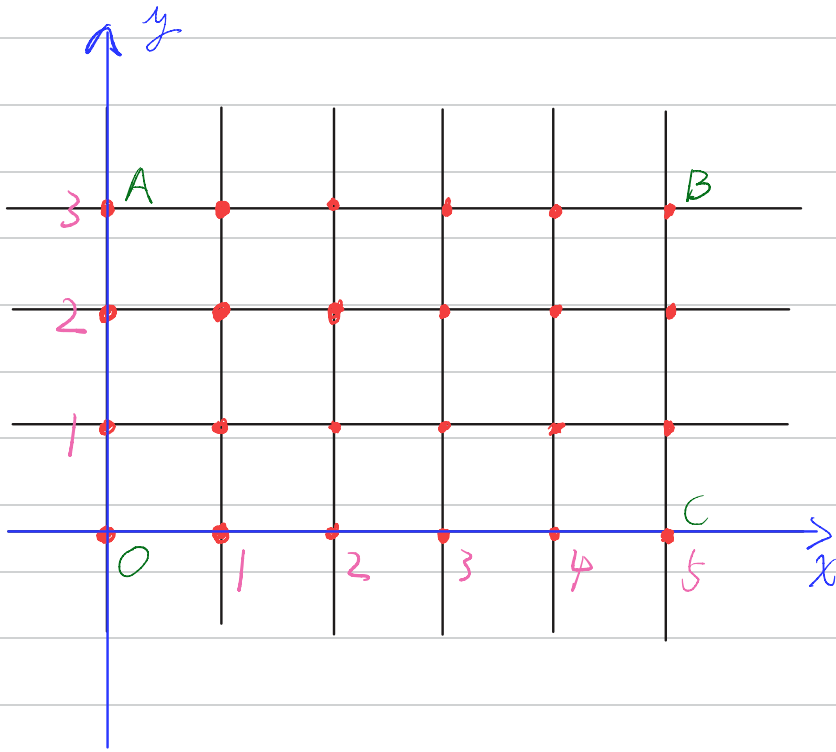
Equivalently,

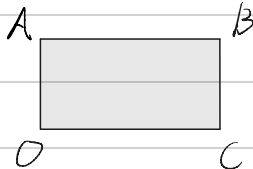
$$\left(\frac{q}{p}\right) = \left(\frac{p}{q}\right),$$

unless $p, q \equiv 3 \pmod{4}$, in which case,

$$\left(\frac{q}{p}\right) = -\left(\frac{p}{q}\right).$$

1. Lattice points in a rectangle



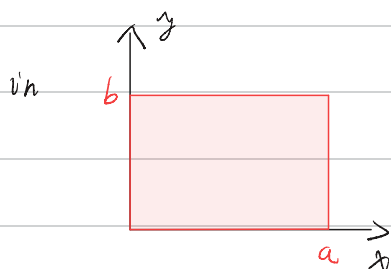
Q1: How many lattice points in  , including

points on AB, BC but not those on OC, OA?

$$A: 15 = 5 \cdot 3 = \text{Area} \left(\begin{array}{c} A \\ \square \\ O \end{array} \begin{array}{c} B \\ \\ C \end{array} \right) : \quad \bullet \iff \square \rightarrow \text{Area 1}$$

Let a, b be positive integers.

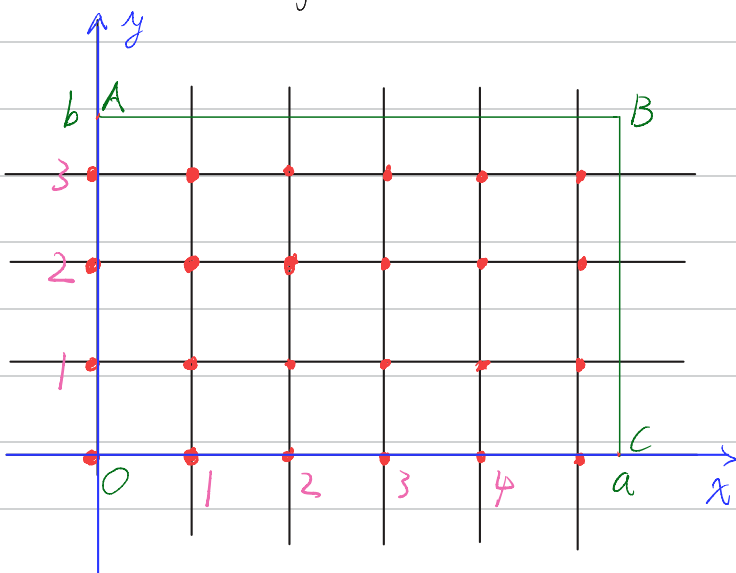
Number of lattice points



$$= \text{Area} = ab.$$

excluding those on the axes

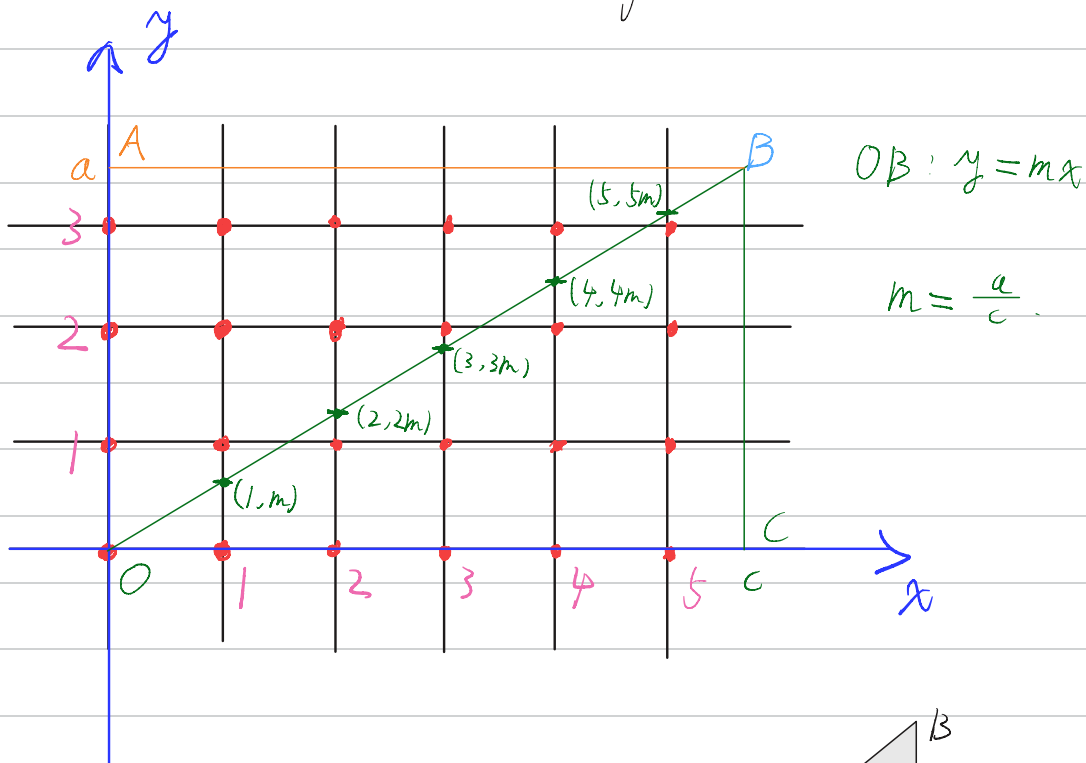
Q2: What if $a, b > 0$ are real numbers?



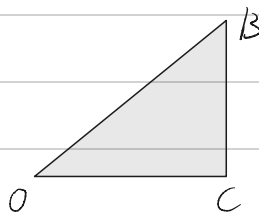
Number of lattice points

$$= \lfloor a \rfloor \cdot \lfloor b \rfloor.$$

2. Lattice points in a triangle



Q1: How many lattice points in

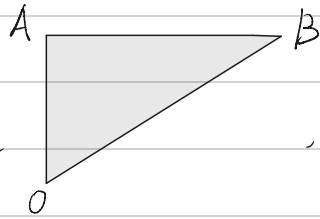


, including

points on OB, BC but not those on OC?

$$A: \lfloor 1 \cdot m \rfloor + \lfloor 2 \cdot m \rfloor + \lfloor 3 \cdot m \rfloor + \dots + \lfloor \lfloor c \rfloor m \rfloor = \sum_{k=1}^{\lfloor c \rfloor} \left\lfloor \frac{ka}{c} \right\rfloor$$

Q2: How many lattice points in , including



points on OB, AB but not those on OA?

$$A: \lfloor 1 \cdot \frac{1}{m} \rfloor + \lfloor 2 \cdot \frac{1}{m} \rfloor + \dots + \lfloor |a| \cdot \frac{1}{m} \rfloor = \sum_{k=1}^{|a|} \lfloor k \cdot \frac{1}{m} \rfloor = \sum_{k=1}^{|a|} \lfloor \frac{kc}{a} \rfloor$$

using OB: $x = \frac{1}{m} \cdot y$.

Eisenstein's proof of quadratic reciprocity:

By Gauss's lemma,

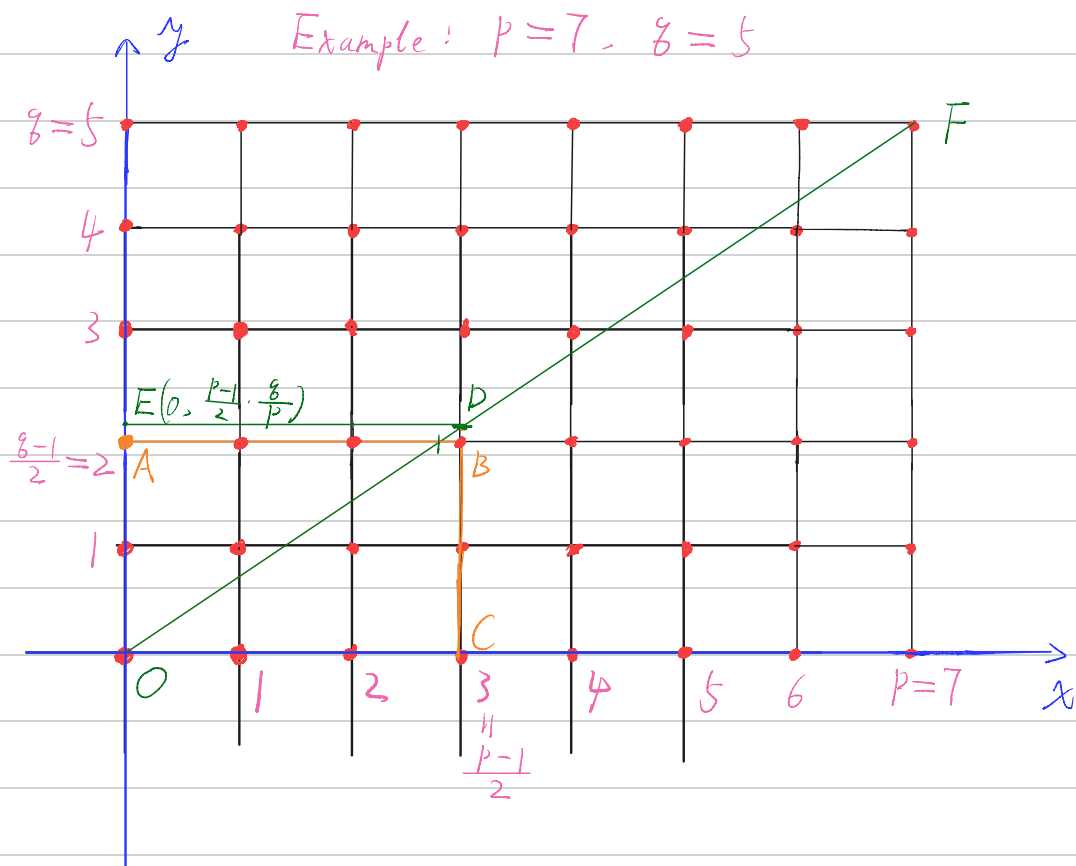
$$\left(\frac{p}{q}\right) = (-1)^{S(p,q)}, \quad S(p,q) = \sum_{k=1}^{\frac{q-1}{2}} \left\lfloor \frac{kp}{q} \right\rfloor.$$

$$\left(\frac{q}{p}\right) = (-1)^{S(q,p)}, \quad S(q,p) = \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{kq}{p} \right\rfloor.$$

$$\Rightarrow \left(\frac{p}{q}\right) \left(\frac{q}{p}\right) = (-1)^{S(p,q) + S(q,p)}$$

$$\stackrel{?}{=} (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}.$$

Need to show $S(p,q) + S(q,p) = \frac{p-1}{2} \cdot \frac{q-1}{2}$.



Line OF : $y = \frac{g}{p} \cdot x$.

Claim: There are no lattice points on OD .

For $1 \leq x \leq \frac{p-1}{2}$, $y = \frac{g}{p} \cdot x \notin \mathbb{Z}$, since $p \neq g \Rightarrow$

$$(p, g) = 1 \text{ and } 1 \leq x < p \Rightarrow p \nmid x, \text{ i.e., } (p, x) = 1.$$

$$D = \left(\frac{p-1}{2}, \underbrace{\frac{p-1}{2} \cdot \frac{g}{p}} \right).$$

$$\notin \mathbb{Z}, \text{ as } p \nmid g, p \nmid (p-1)$$

Also, since $p > g$, we have

$$\frac{g-1}{2} < \frac{p-1}{2} \cdot \frac{g}{p} < \frac{g+1}{2}.$$

$$\Rightarrow \left\lfloor \frac{p-1}{2} \cdot \frac{g}{p} \right\rfloor = \frac{g-1}{2}.$$

$$\text{Number of lattice points in } \begin{array}{c} E \\ \square \\ O \quad C \end{array}^D = \frac{p-1}{2} \cdot \frac{g-1}{2}.$$

//

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$$\text{Number of lattice points in } \begin{array}{c} \triangle \\ O \quad C \end{array}^D = \sum_{k=1}^{\frac{p-1}{2}} \left\lfloor \frac{kg}{p} \right\rfloor = S(g, p)$$

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$$\text{Number of lattice points in } \begin{array}{c} E \\ \triangle \\ O \end{array}^D = \sum_{k=1}^{\frac{g-1}{2}} \left\lfloor \frac{kp}{g} \right\rfloor = S(p, g)$$

$$\text{Eg } \left(\frac{90}{103} \right) = \left(\frac{2 \cdot 3^2 \cdot 5}{103} \right) = \left(\frac{2}{103} \right) \left(\frac{3}{103} \right)^2 \left(\frac{5}{103} \right)$$

$\uparrow$ prime $\parallel$
 $\parallel$

$$103 = 8 \cdot 12 + 7 \equiv 7 \pmod{8} \Rightarrow \left(\frac{2}{103} \right) = 1.$$

$$5 \equiv 1 \pmod{4}, \quad 103 \equiv 3 \pmod{4}$$

$$\text{QR} \Rightarrow \left(\frac{5}{103} \right) = \left(\frac{103}{5} \right) = \left(\frac{3}{5} \right) = \left(\frac{5}{3} \right) = \left(\frac{2}{3} \right) = -1.$$

$\parallel$
 $1 \pmod{4}$

$$\text{So } \left(\frac{90}{103} \right) = -1.$$

Jacobi symbol.

Given $a \in \mathbb{Z}$ and a positive odd integer $n = p_1^{m_1} \cdots p_k^{m_k}$,

define

$$\left(\frac{a}{n}\right) = \left(\frac{a}{p_1}\right)^{n_1} \left(\frac{a}{p_2}\right)^{n_2} \cdots \left(\frac{a}{p_k}\right)^{n_k},$$

with $\left(\frac{a}{1}\right) = 1$.

Observation :

$$\left(\frac{a}{n}\right) = \begin{cases} \pm 1 & , (a, n) = 1, \\ 0 & , \text{otherwise} \end{cases}$$

Properties : Let $a, b \in \mathbb{Z}$ and $m, n \geq 1$ be odd integers.

1. (Periodicity) If $a \equiv b \pmod{n}$, then

$$\left(\frac{a}{n}\right) = \left(\frac{b}{n}\right).$$

2. (Complete multiplicativity)

$$\left(\frac{ab}{n}\right) = \left(\frac{a}{n}\right)\left(\frac{b}{n}\right).$$

$$\left(\frac{a}{mn}\right) = \left(\frac{a}{m}\right)\left(\frac{a}{n}\right).$$

$$(3) (-1 \bmod n) \left(\frac{-1}{n}\right) = (-1)^{\frac{n-1}{2}} = \begin{cases} 1, & n \equiv 1 \pmod{4} \\ -1, & n \equiv 3 \pmod{4} \end{cases}$$

$$(4) (2 \bmod n) \left(\frac{2}{n}\right) = (-1)^{\frac{n^2-1}{8}} = \begin{cases} 1, & n \equiv 1, 7 \pmod{8} \\ -1, & n \equiv 3, 5 \pmod{8} \end{cases}$$

(5) (Quadratic reciprocity) If $(m, n) = 1$, then

$$\begin{aligned} \left(\frac{m}{n}\right)\left(\frac{n}{m}\right) &= (-1)^{\frac{m-1}{2} \cdot \frac{n-1}{2}} \\ &= \begin{cases} -1, & m, n \equiv 3 \pmod{4} \\ 1, & \text{otherwise.} \end{cases} \end{aligned}$$

Pf. HW 2.

Warning! If $a \in \mathbb{Z}$ is a quadratic residue n in the

sense that $(a, n) = 1$ and there is $x \in \mathbb{Z}$ s.t.

$$x^2 \equiv a \pmod{n},$$

then $\left(\frac{a}{n}\right) = \left(\frac{x^2}{n}\right) = \left(\frac{x}{n}\right)^2 = 1$, as $(a, n) = 1 \Rightarrow (x, n) = 1$.

But the converse is false. For instance,

$$\left(\frac{5}{9}\right) = \left(\frac{5}{3}\right)^2 = 1,$$

but there is no $x \in \mathbb{Z}$ s.t. $x^2 \equiv 5 \pmod{9}$.

Sum of two squares

Q: Which positive integers n can be expressed as

$$n = x^2 + y^2, \quad x, y \in \mathbb{Z} ?$$

n	$\square + \square$	n	$\square + \square$
1	$1^2 + 0^2$	13	$2^2 + 3^2$
2	$1^2 + 1^2$	14	X
3	X	15	X
4	$2^2 + 0^2$	16	$4^2 + 0^2$
5	$1^2 + 2^2$	17	$1^2 + 4^2$
6	X	18	$3^2 + 3^2$
7	X	19	X
8	$2^2 + 2^2$	20	$2^2 + 4^2$
9	$3^2 + 0^2$	21	X
10	$1^2 + 3^2$	22	X
11	X	23	X
12	X	24	X